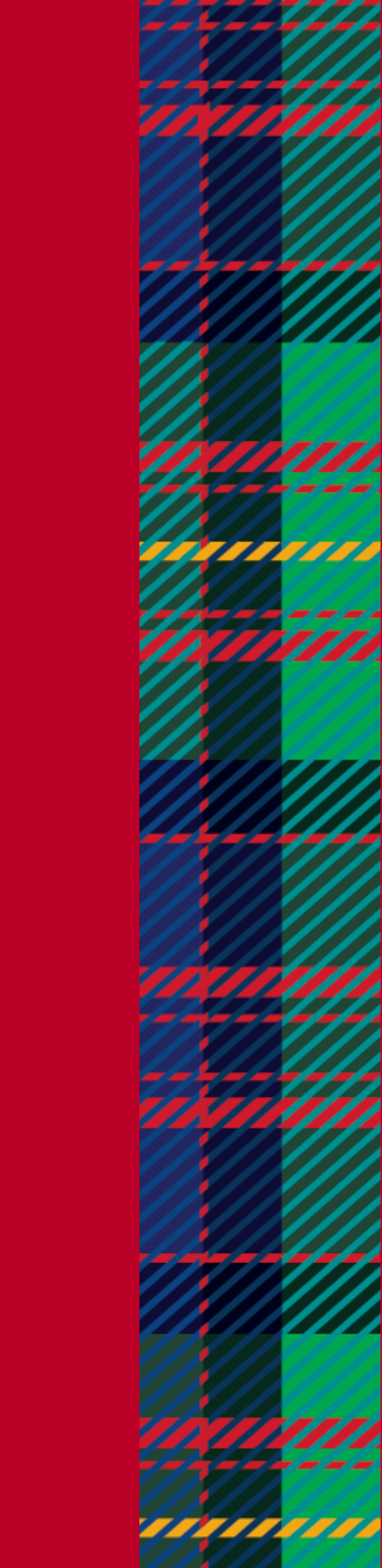




Carnegie Mellon University

FPN-Mamba: A Feature Pyramid Network with Bidirectional State Space Models for Medulloblastoma Classification

Adetoro Damilola | Adeniji Samuel

22/04/2026

INTRODUCTION

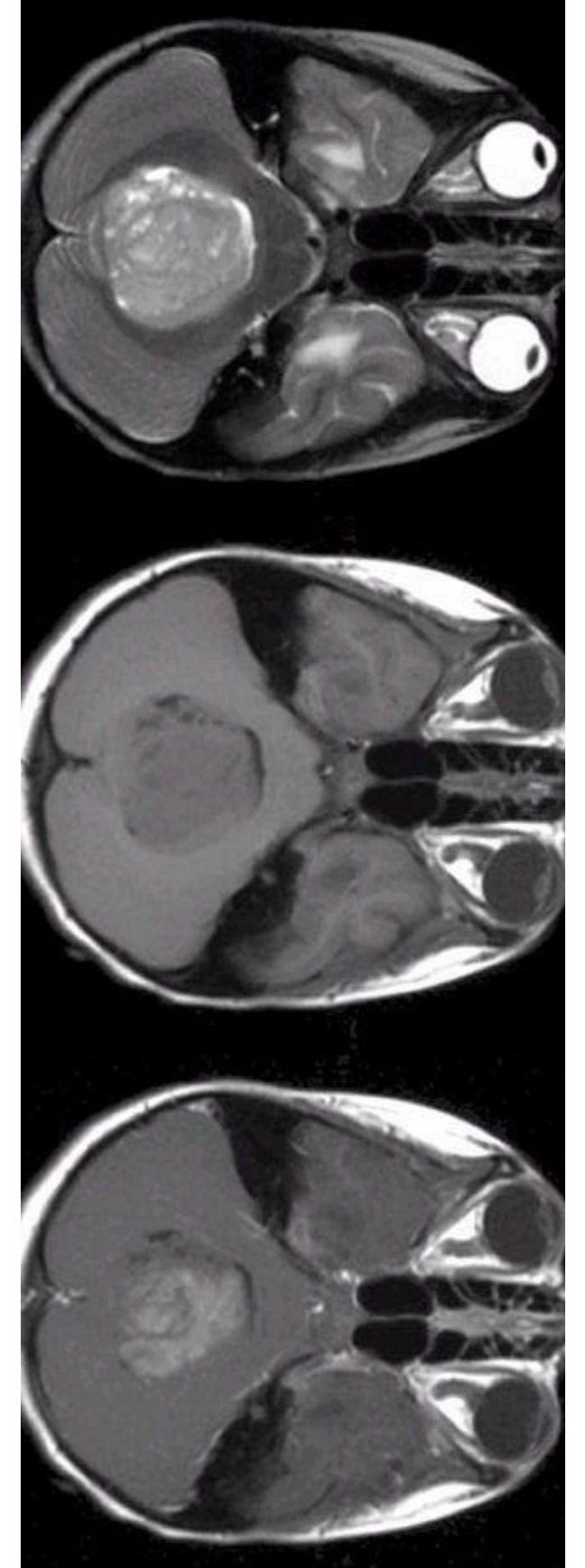
The Clinical Problem

20%

of pediatric brain tumors

*A missed case is not a metric error.
It is a child without a diagnosis.*

- Most common malignant brain tumor in children
- Missed diagnosis = delayed treatment, worse outcomes
- Hard to detect: small lesions, low contrast, similar appearance to other tumors
- MRI is the primary tool, but interpretation is difficult even for experts



RELATED WORKS

Method	Multi-scale	Global context	Complexity
CNN + attention Rasheed 2024, Apostolopoulos 2023	No	No	$O(L^2)$
FPN Lin et al. 2017	Yes	No	$O(L)$
Mamba SSM Gu & Dao 2023	No	Yes	$O(L)$
InceptionNet Fang et al. 2025, our baseline	Partial	Single scale	$O(L^2)$
FPN-Mamba (ours) classification	Yes	Yes	$O(L)$

Gap: no work combines both for MB classification

RELATED WORKS

Why Improve a Strong Baseline?

InceptionNet

Fang et al. 2025

98.07%

Accuracy

99.41%

AUC

96.03%

Recall

Single-scale limitation

Attention applied to one coarse feature map small, subtle lesions are structurally invisible

False negatives cluster here

Fang et al. admit FNs concentrate on small lesions, low contrast, atypical location

Goal: clinical, not benchmarking

Reducing FN rate from ~500 to 300 means fewer children misdiagnosed

Dataset and Setup

761

total images

131 vs 630

MB vs Non-MB

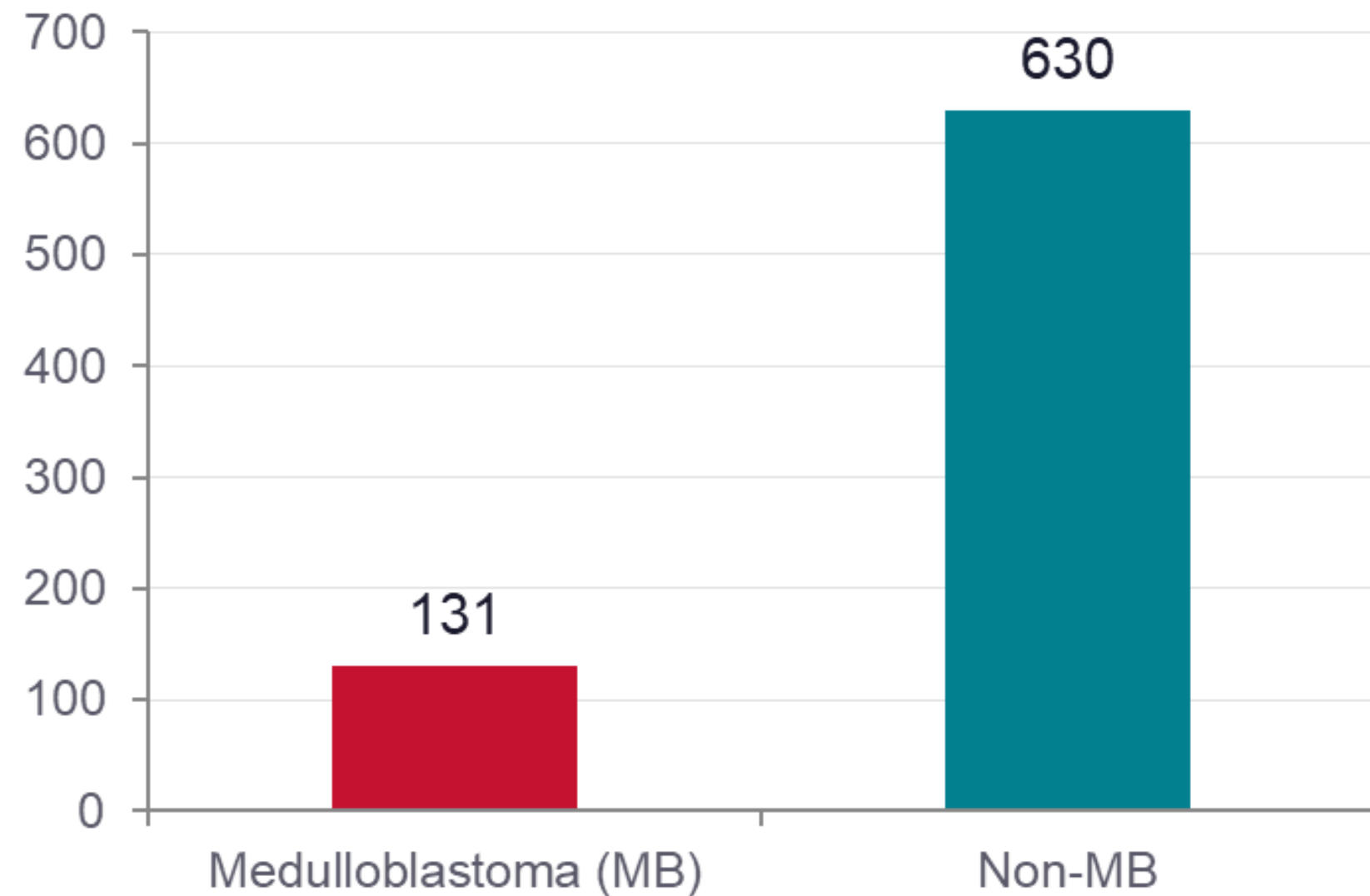
5-fold

stratified CV

4.85×

pos_weight correction

Class Distribution



Heavier augmentation applied to MB class only to address imbalance

Architecture: FPN-Mamba



Multi-scale

Each pyramid level sees the tumor at a different resolution — small lesions are not lost

$O(L)$ complexity

Mamba scales linearly — global context is feasible at fine-resolution levels unlike $O(L^2)$ attention

LocalityMixing

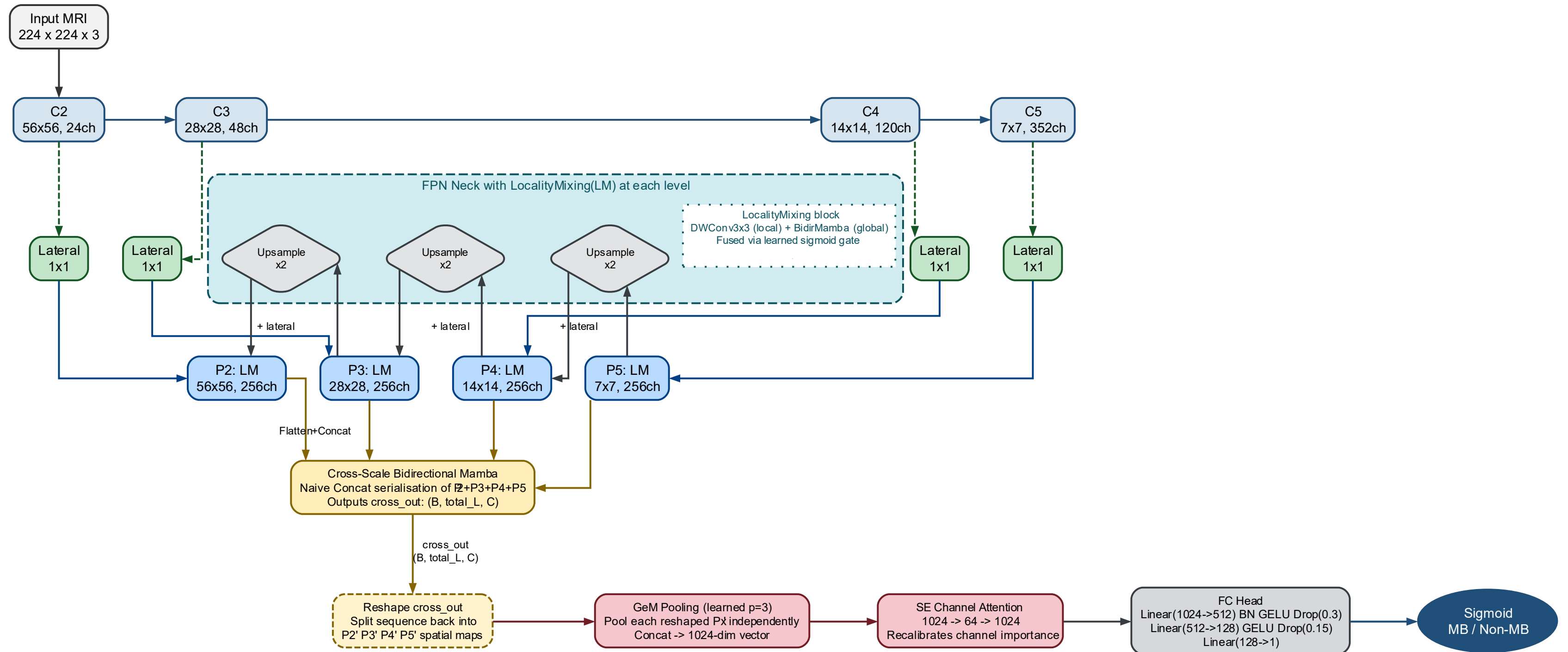
DWConv for local detail + Bidirectional Mamba for global context, blended by a learned gate

GeM + SE

Emphasises high-activation regions; SE recalibrates which channels matter most before classification

METHODOLOGY

FPN-Mamba Architecture



Quantitative Comparison

4 of 5

metrics beat baseline

+2.80pp

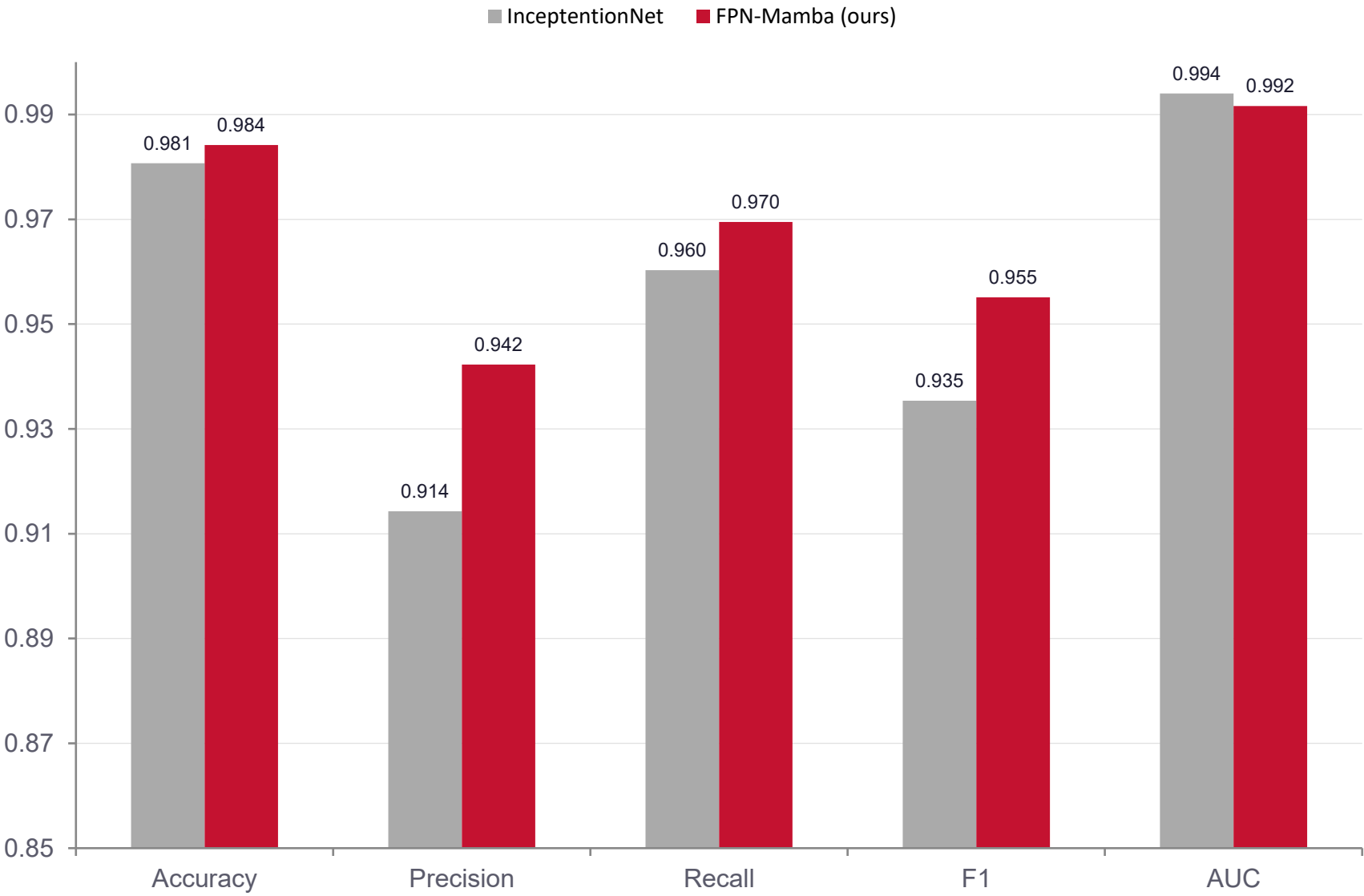
precision improvement

+1.97pp

F1 improvement

3 FN

40% decrease in recall



Sensitivity/Recall 0.970, Specificity: 0.966

Summary and Future Work

Multi-scale detection

FPN ensures tumors at every size are represented. Nothing is discarded before the classifier sees it.

Linear-complexity context

Mamba provides global context at fine-resolution levels — where small tumors live — without quadratic cost.

Clinical objective

3 false negatives vs ~5 baseline. Metrics are a byproduct. Fewer missed diagnoses is the goal.

Limitations: No public external MB dataset · No formal ablation study completed

Next: Grad-CAM at each pyramid level · Formal ablation study · Larger institutional dataset

Concepts Learnt

Feature Pyramids in practice

The FPN concept taught in class directly shaped the architecture, seeing it applied to real medical imaging showed how powerful multi-scale thinking is beyond object detection.

Research requires depth

Extensive literature review led to better design choices. Understanding the baseline's limitations was as important as building our own solution.

Beyond the benchmark

Research in healthcare is not about improving a score. Every false negative has a human cost. That reframing changed how we thought about every design decision.

REFERENCES

Apostolopoulos, I.D., Aznaouridis, S., Tzani, M.: An attention-based deep convolutional neural network for brain tumor and disorder classification and grading in MRI. *Information* 14(3), 174 (2023)

Rasheed, Z., Ma, Y.K., Ullah, I., Al-Khasawneh, M., Almutairi, S.S., Abohashrh, M.: Integrating convolutional neural network with attention mechanisms for MRI-based classification of brain tumors. *Bioengineering* 11(7), 701 (2024)

Lin, T.Y., Dollar, P., Girshick, R., He, K., Hariharan, B., Belongie, S.: Feature pyramid networks for object detection. In: *2017 IEEE Conference on Computer Vision and Pattern Recognition*, pp. 936–944. IEEE, Honolulu (2017)

Gu, A., Dao, T.: Mamba: Linear-time sequence modeling with selective state spaces. *arXiv:2312.00752* (2023)

Fang, C., Li, C., Liu, H., Zhou, Q., Li, S., Chen, H., Chen, X., Shen, Z.: Precise identification of medulloblastoma in MRI using a convolutional neural network integrated with a self-attention mechanism. *Digit. Health* 11, 20552076251351536 (2025)



HALL TOWER



MELLON
INSTITUTE

Andrew Carnegie



CARNIVAL



Carnegie
Mellon
University



SWEEPSTAKES

FINISH

START



FOUNDED
1900



BOOTH
OF MIDWAY



BE THE
FIRST
PENGUIN

— Randy Pausch



MOBOT
FINALS



*Cardus Marie,
Ladies Thistle.*



*Diplodocus
Carnegii*